Crop Price Forecasting using ML and Flask

1. Algorithm

1. Input Data Preparation:
   - Load historical crop price data (CSV).
   - Preprocess data: features = [Month, Year, Rainfall], target = [WPI].

2. Model Training:
   - Train a Decision Tree Regressor for each crop using scikit-learn.
   - Use a random max_depth between 7-18 for model variation.

3. Prediction:
   - Predict WPI for a given crop/month/year/rainfall.
   - Convert WPI to price using predefined base price table.

4. Forecasting:
   - Forecast next 6 and 12 months prices using trained model.
   - Calculate % change from current month for visualization.

5. Web Integration:
   - Use Flask to build web endpoints and integrate the model.
   - Render results with Jinja2 templates in HTML.

2. Methodology

- Collect and store crop price data in CSV files.
- Train individual ML models using Decision Tree Regressor.
- Build a Flask web app to serve predictions and analytics.
- Forecast and visualize trends using rainfall, month, and year as inputs.
- Sort and display top 5 winners and losers by predicted price change.

3. Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: 4 GB minimum
- Storage: 1 GB free space
- Display: Minimum 1024x768 resolution
- GPU: Not mandatory

4. Software Requirements

- OS: Windows, Linux, or macOS
- Python: Version 3.6 or higher
- Libraries: flask, pandas, numpy, scikit-learn, flask_cors
- IDE: VSCode / PyCharm
- Web Browser: Chrome / Firefox
- Web Framework: Flask